22/07/26
Wednesday

Assignment - I

Challa priyadarshini
192311118
Cryptography & Network Security

Question 1

- Encrypt the message "DEFEND THE WALL" using caesar cipher with the key = 3.

→ shift each letter by +3.

D → G

PT:	D	E	F	E	N	D	T	H	E	W	A	L	L
CT:	G	H	I	H	Q	G	W	K	H	Z	D	O	O

cipherText : GHIHQG WKH ZDOO

100
100

- Decrypt "KHOOR ZRUOG" using caesar cipher with key = 3

shift each letter - 3.

Cipher Text :	K	H	O	O	R	Z	R	U	O	G
plain Text :	H	E	L	L	O	W	O	R	L	D

plainText: HELLOWORLD

- A caesar cipher is used to encrypt the Word "SECRET" to "VHFUHW". Determine the key used, and decrypt "WKLV LV D WHVW"

→ Given SECRET → VHFUHW

Each letter shifted by +3

K = 3

Now decrypt WKLVLVD WHVW

CT	W	K	L	V	L	V	D	W	H	V	W
PT	T	H	I	S	I	S	A	T	E	S	T

Question 2

- using the substitution key:

A → Q, B → W, C → E, D → R, E → T, F → Y ... Z → M

Encrypt the word 'FIRE'.

F	I	R	E
Y	O	K	T

- Given this cipher text "UJHHD QEB NRFZH YOLTK" and knowing it is encrypted with a monoalphabetic substitution, find the plainText using frequency analysis.

CT	U	J	H	H	D	Q	E	B	N	R	F	Z	H	Y	O	L	T	K
PT	H	E	L	L	O	T	H	E	W	O	R	I	D	A	G	A	I	N

Question 3

- Encrypt the message "ATTACKATPALACE" using the keyword 'LEMON'.

plainText: ATTACK AT PALACE

keyword: LEMON LEMON LEMON

using vigneee cipher.

Cipher Text :- LXFOPVEFDNWIEOS

- Decrypt the message "LXFOPVEFRNHR" using the keyword "LEMON"

CipherText: LXFOPVEFRNHR

keyword: LEMON

plainText: ATTACK DAWN

- Given the encrypted message "QNIXEPKMAEGTU" using the vignere cipher and keyword "KEY". Find the original message.

Keyword : key

Repeated key: Key Key Key Key

After substituting key values,

original message : CRYPTOGRAPHY

Question 4:

- Given a 2x2 key matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Encrypt the message "HI" using the Hill cipher

Given PT: H I

$\downarrow$ $\downarrow$

7 8

Multiply

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{matrix} 3(7) + 3(8) = 45 \\ 2(7) + 5(8) = 54 \end{matrix}$$

modulo 26

45 mod (26) = 19 = T

54 mod (26) = 2 = C

cipher Text : TC

- If the message "PO" was encrypted to "QC" using the above HILL CIPHER matrix, verify the result by showing the matrix multiplication

Given P O

15 14

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \end{bmatrix} = \begin{bmatrix} 87 \\ 100 \end{bmatrix}$$

87 mod(26) = 7 = J

100 mod (26) = 22 = W

Result is JW, not QC.

Therefore po → QC is incorrect for this Matrix.

Question 5:

- Encrypt the message "MEETMEAFTERLUNCH" using "ZEBRAS"

plainText: MEETMEAFTERLUNCH

Keyword: ZEBRAS

Repeated: ZEBRASZEBRASZEB

using vigneee encryption

cipheeText:- MRXETCEFNTEHELXMAU

- Decrypt the message "EANMTEFCUHETREL" using "BOARD"

cipheeText: EANMTEFCUHETREL

Keyword: BOARD

plainText: SECRET MESSAGE

Question 6:

- using the one-time pad key XMCKL, encrypt the message "HELLO"

Key: XMCKL

plaintText: "HELLO"

H	E	L	L	O	X	M	C	K	L
7	4	11	11	14	23	12	2	10	11

$7+23 = 30 \bmod 26 = 4$
$4+12 = 16 \bmod 26 = 16$
$11+2 = 13 \bmod 26 = 13$
$11+10 = 21 \bmod 26 = 21$
$14+11 = 25 \bmod 26 = 25$

cipheeText:
EQNVZ

Q. Explain why the message "ATTACKNOW" encrypted using OTP with key "QWGRTYUIO" is unbreakable under ideal conditions.

- => The key is completely random
- => The key length equal the message length
- => The key is used only once
- => The key is kept secret.

Question 7

- Encrypt the message:

  PT: "WE ARE DISCOVERED"

  KEYWORD: "ZEBRA".

1. Arrange the Text

Z	E	B	R	A
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

A B E R Z

A → 5 B → 3 E → 2 R → 4 Z → 1

A : EOD

B : ASR

E : EIE

R : RCE

Z : WDV

Cipher Text :- EODASREEICERCWDV

- Decrypt the message:

  CT: "EVLNEACDTKESEAQROFOIDGECUXWN"

  Key: "TRIAL"

alphabetic ~~ord~~

WE ARE DISCOVERED FLEE AT ONCE

- The message "INFORMATIONSECURITY" is encrypted with "HACK"

$L = 19$ Row needed : 5

$c = 4$ padding required : 1x

A → NMOCT C → FANUY H → IRIEI K → OTSRX

NMOCTFANUYIRIEIOTSRX

8. you are given the following CT encrypted text using a monoalphab[etic]
substitution cipher. UZQSOVUOHX MOPVGPVG POZPEVSGZUIS ZOPFPG SX
UZUETUCG.

Cipher text

letter freq

letter	count
Z	7
U	5
O	5
P	4
S	4
V	3
E	3

Since E is the most common letter in english

Z → E   U → T   O → A   P → O   S → I

9. you intercepted a message encrypted with a monophabetic substitution cipher.

CT: BJJWI QJWI YMJWI NXF XNRUQJWX

Original plainText with "THIS IS THE".

Mapping:

T → B   H → J   I → WI   S → Q   E → M

continue Decryption

This is the ?? SIMPLE?

plainText:

THIS IS THE SIMPLEST

You are given the cipher text below encrypted using the Vigneere cipher with an unknown short key (2-5 letter long):

CT: ZJCTWQNGRZGVTWIDVZHCOYGIMGJ

S-1: Key length

Repeated pattern suggest a 3-letter key.

S-2: Recover key

frequency analysis

S-3: Decrypt

THISISAVIGENERCIPHERMESSAGE

Explaination:

* Recover the key using frequency analysis.
* Decrypt each letter by subtracting the key value mod26.